

FFT-Based Vibration and Power Spectral Density Analysis

of IMU Data, with In-Flight Motor RPM Synchronization on FPV Drone

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What an IMU Measures and Why a Drone Needs One

What is an IMU?

An Inertial Measurement Unit is a MEMS sensor package that measures the motion of its own body — without any external reference.

Accelerometer:

Linear acceleration along three orthogonal axes, in units of g. At rest it reads the gravity vector (≈ 1 g vertical), which lets it sense tilt.

Gyroscope:

Angular rate around the same three axes, in rad/s or $^{\circ}/s$. It tells how fast the body is rotating, not how it is oriented.

Why a quadcopter needs it

- A quadcopter is mechanically unstable — without active control it cannot stay level.
- The flight controller runs a PID loop hundreds of times per second to hold attitude. The PID needs to know how the airframe is moving right now.
- Gyroscope rates are the direct error signal of the inner PID loop.
- Accelerometer is fused with the gyro to correct long-term drift and recover absolute orientation.
- If the IMU signal is noisy, the controller fights the noise instead of the real motion.

The IMU is the only sensor that closes the attitude control loop — its quality directly limits flight performance.

From Raw IMU Signals to the Frequency Domain

The motion of a flying drone contains many superimposed oscillations: motor rotation, blade-pass effects, frame vibration, and the pilot's actual commands. The time-domain signal looks like noise; the frequency domain separates the contributors.

What the Fourier transform does

- Decomposes a time signal into the sinusoidal components it is built from.
- Reveals at which frequencies the energy concentrates — peaks in the spectrum correspond to periodic physical processes.
- Welch's PSD averages many short FFTs to give a stable estimate of vibration energy versus frequency.
- A spectrogram is the same idea over time — useful when the dominant frequency is not constant.

How we use it on this drone

- Apply Welch PSD and the spectrogram to the logged 1 kHz gyro and accel signals.
- Identify peaks driven by motor rotation ($f_0 = \text{RPM} / 60$) and blade-pass ($2 f_0$ for two-blade, $3 f_0$ for three-blade).
- Overlay the live RPM signal on the spectrogram to attribute each peak to a specific motor state.
- Use the result to characterize the vibration spectrum and inform dynamic notch filter design.

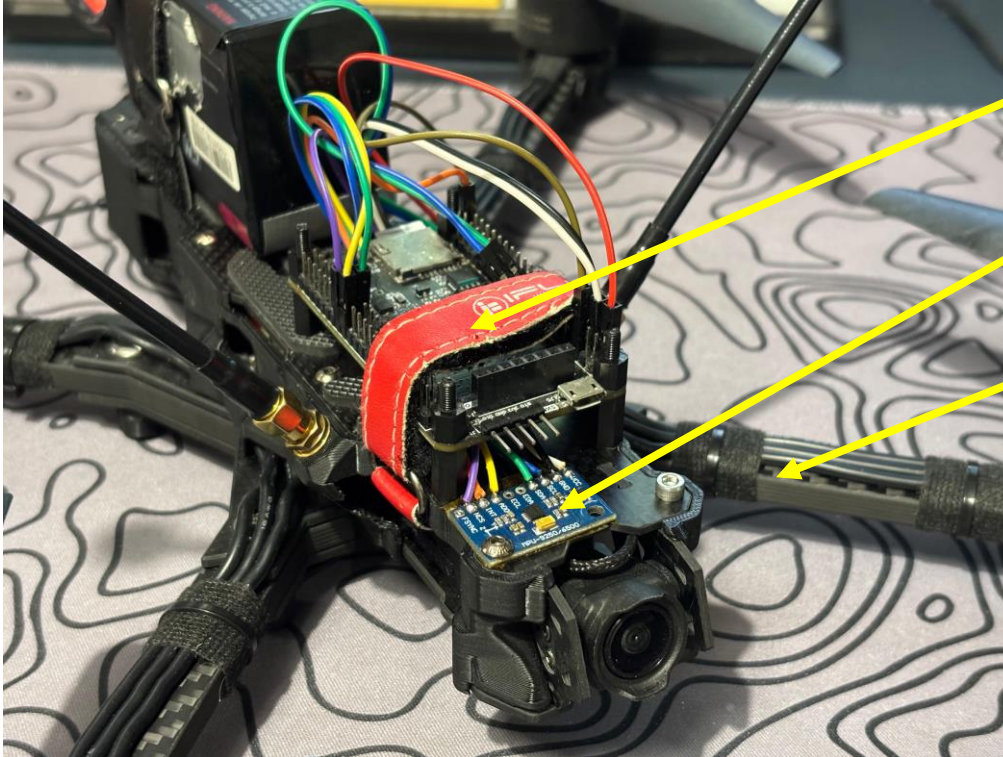
Fourier analysis turns an unreadable time series into the engineering quantity we actually need: where the vibration energy is.

Why Vibration Analysis Matters on a Drone

- FPV drones suffer from motor- and propeller-induced vibration that contaminates the gyroscope signal used by the flight controller.
- Vibration energy scales roughly with the square of motor RPM, so its dominant frequencies shift with throttle.
- If unfiltered, this noise degrades the PID inner loop, attitude estimation, and ultimately flight stability.
- Modern flight controllers use dynamic notch filters tuned to the live motor RPM — but designing one requires data.

Goal: capture high-rate IMU data and motor RPM simultaneously, then quantify the vibration spectrum experimentally.

Measurement Platform



MCU / logger

STM32H7 Microcontroller with SD Card for sensor data logging.

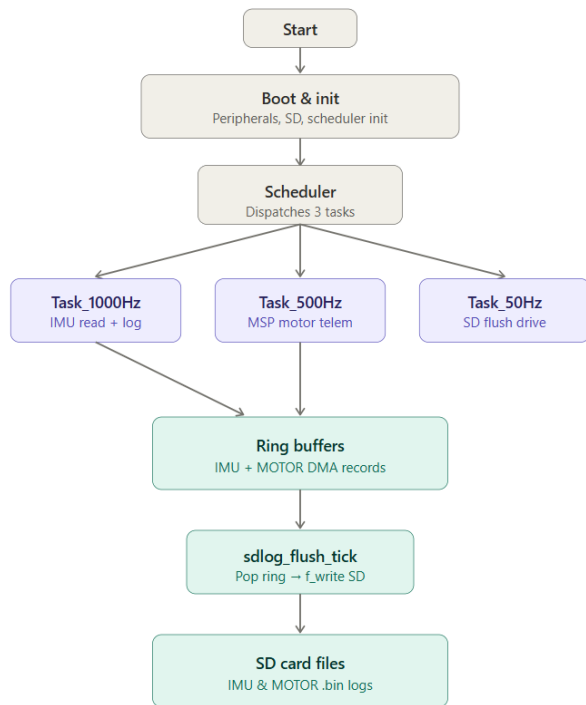
IMU

MPU9250 over SPI, 1 kHz accel + gyro sampling.

Test Platform

10 inch drone with 6S battery (24V, 8 kg peak thrust)

STM32 Data Flow — From Sensors to SD Card



1 kHz Sampling with Microsecond Timestamps

- MPU9250 in SPI mode; sample-ready interrupt drives a non-blocking DMA read in Task_1000Hz.
- Internal DLPF set to 92 Hz cutoff, gyro full-scale $2000^{\circ}/s$, accel ± 16 g.
- Timestamps from DWT cycle counter with software 64-bit extension — microsecond resolution, shared with the motor log.
- Per-axis bias and scale calibration loaded from flash at boot.
- Each sample: 32-byte packed record (uint64 ts_us + 6 float32 channels) pushed to the IMU ring buffer.
- Effective achieved rate ≈ 980.5 Hz; the $\sim 1.75\%$ drop is the SD card flush window, not the MCU.

Binary SD Logging with Multi-Slot Ring Buffers

- Custom sdlog library: each data stream is a slot with its own header, ring buffer, and flush policy.
- 32-byte SDLB header per file: magic, version, record size, rate, slot name — self-describing.
- Ring buffer placed in non-cacheable AXI SRAM (MPU Region 0) so DMA writes do not need cache maintenance.
- Flush task runs at 50 Hz, bounded to 4 KB per tick to avoid blocking real-time loops.
- Two parallel slots used: IMU ($1 \text{ kHz} \times 32 \text{ B} \approx 32 \text{ KB/s}$) and MOTOR ($50 \text{ Hz} \times 64 \text{ B} = 3.2 \text{ KB/s}$).
- Byte-perfect on-disk integrity verified across hundreds of files.

Aligning 50 Hz Motor Data onto the 1 kHz IMU Timeline

- Both logs share the same `get_us_64()` microsecond counter — no clock-drift correction is needed.
- Motor records are linearly interpolated onto each IMU sample to produce a per-sample (RPM, throttle) tuple at 1 kHz.
- Smart zero-fill: short zero-RPM runs (≤ 100 ms, typically transient DSHOT dropouts) are interpolated; long runs (disarmed / idle) are preserved as motor-off.
- A validity mask marks which samples are real measurements vs. interpolated, so downstream metrics can distinguish them.
- Flight-only mask: samples with max-throttle below $1100 \mu\text{s}$ are excluded from PSD averaging to remove disarm silence.

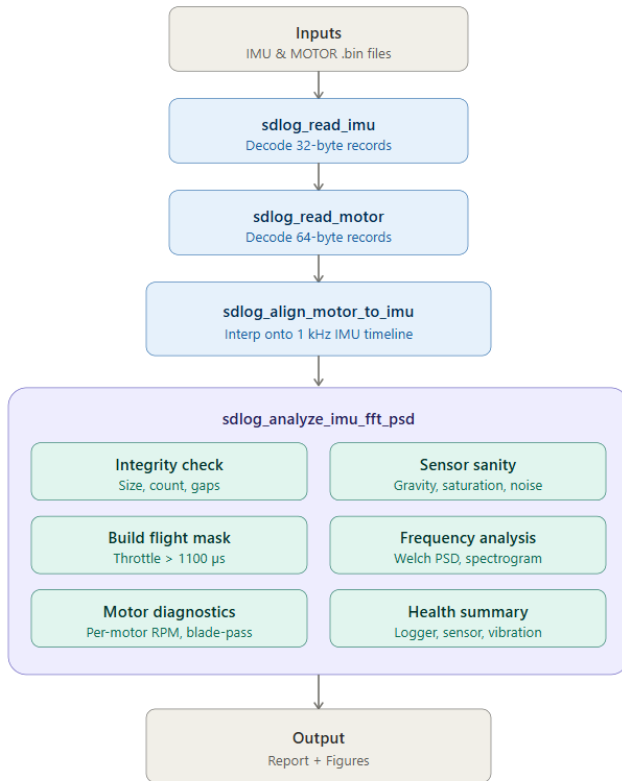
Result: the analysis can answer "what was the motor doing at the exact moment this vibration peak occurred?"

From Raw Binary Logs to Engineering Metrics

Input	Processing stages	Output
IMU_1000Hz_LOG_NNN.bin MOTOR_100Hz_LOG_NNN.bin	Parse binary header + records Integrity check (rate, gaps) Motor → IMU alignment + zero-fill Flight-only PSD masking FFT / Welch PSD / Spectrogram Harmonic detection Health scoring	Logger / sensor / vibration scores PSD plots and spectrograms RPM overlay figures Notch frequency range guidance Text report (terminal)

All formulas implemented in MATLAB; binary loaders, alignment, and analysis are modular and reusable.

Per-Log Pipeline



Each stage produces both numerical metrics (returned in the report struct) and figures.

Verifying the Quality of the Logged Data

- Effective sample rate from monotonic timestamps, compared to nominal.
- Gap detection: $dt > 1.5 \times \text{nominal}$ counted as a gap, total samples lost reported.
- Timestamp monotonicity and duplicate detection (boot-time and clock-rollover sanity).
- Saturation check on accel and gyro channels.
- Stationary noise floor measured in the first 5 s window: compared against MPU9250 spec sheet.

Example output (this dataset)

Expected rate: 1000 Hz
Measured rate: 980.5 Hz • Missed samples: 1.75%
Longest gap: 25.4 ms • Non-monotonic / duplicates: 0 / 0

FFT, Welch PSD, and the Spectrogram

Discrete Fourier Transform

$$X[k] = \sum x[n] \cdot \exp(-j \cdot 2\pi \cdot k \cdot n / N), \quad n = 0 \dots N-1$$

Welch Power Spectral Density

Segment the signal into overlapping windows (Hann, 50% overlap), compute $|FFT|^2$ of each, normalize by window energy, then average:

$$P(f) = (1 / K) \cdot \sum_{k=1..K} P_k(f)$$

Short-Time Fourier Transform (Spectrogram)

$$S(t, f) = | \sum x[n] \cdot w[n-t] \cdot \exp(-j \cdot 2\pi \cdot f \cdot n) |^2$$

From Mechanical RPM to Expected Vibration Frequencies

Mechanical fundamental frequency of a motor

$$f_0 = RPM / 60 \quad [Hz]$$

Blade-pass frequency (B-blade propeller)

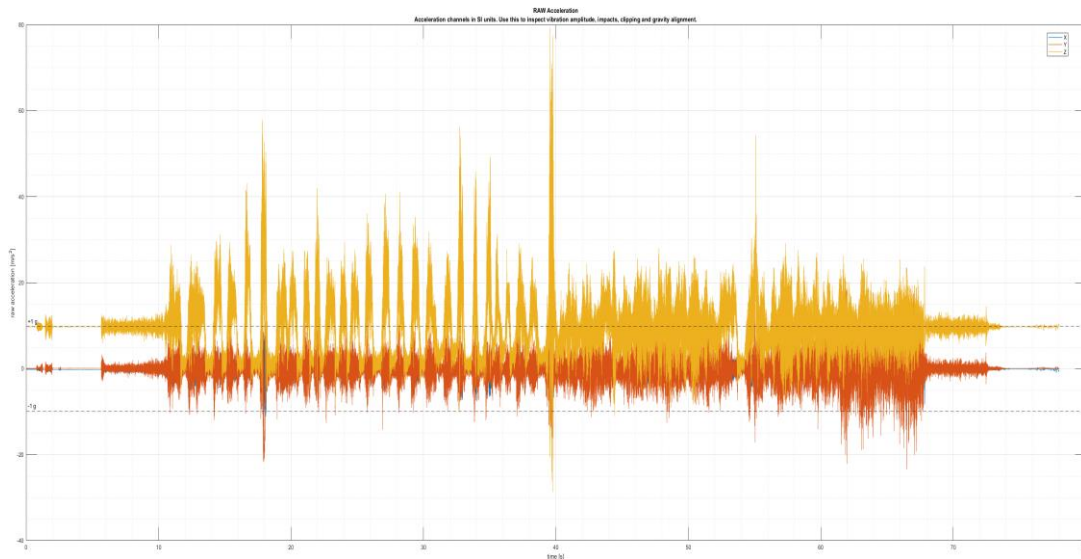
$$BPF = B \cdot f_0 = B \cdot RPM / 60 \quad [Hz]$$

$$\text{Two-blade prop} \Rightarrow 2 f_0 \quad \cdot \quad \text{Three-blade prop} \Rightarrow 3 f_0$$

Vibration peaks classified by the analyzer

- Fundamental f_0 and harmonics $2f_0 \dots 6f_0$
- Blade-pass BPF and harmonics $2 \cdot BPF \dots 6 \cdot BPF$
- Match tolerance $\pm 8\%$ of candidate frequency

Raw Time-Domain Acceleration



Why look at raw data?

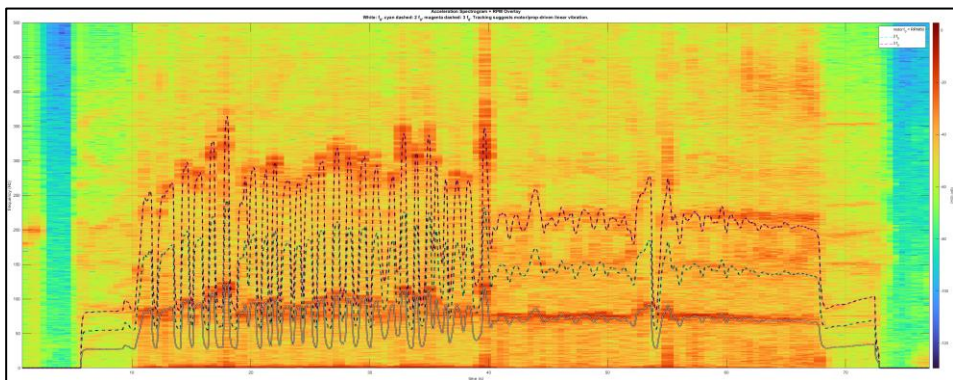
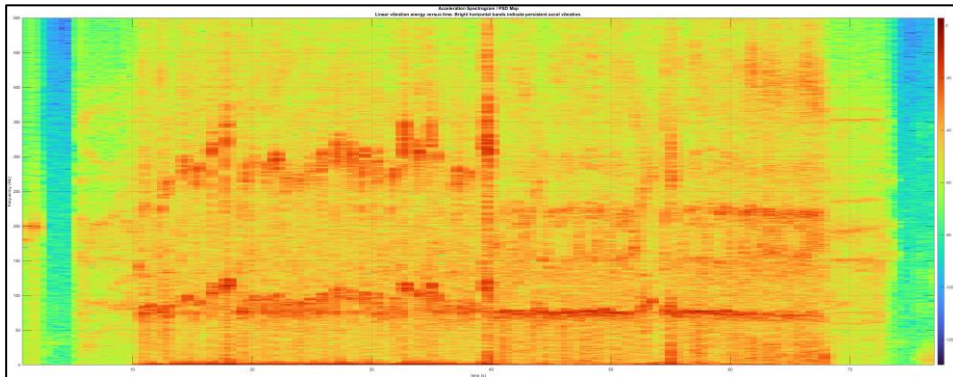
Frequency-domain plots can hide events that the time domain makes obvious.

What this view exposes

- IMU saturation: Flat-top clipping at ± 16 g.
- Mechanical impacts: Sharp spikes from prop strikes or hard landings.
- Vibration envelopes: Slow modulation with throttle and aerodynamic load.
- Resonance build-up: Slow amplitude growth at a fixed frequency.

This view is the first sanity check before any spectral analysis — it answers "is the data physically reasonable?"

Gyro Spectrogram with Live RPM Overlay



Why the gyro?

Direct input to the PID inner loop. Vibration energy here directly corrupts the controller.

Read this plot

White line — $f_0 = \text{RPM} / 60$.

Cyan dashed — $2f_0$ (two-blade prop).

Magenta dashed — $3f_0$ (three-blade prop).

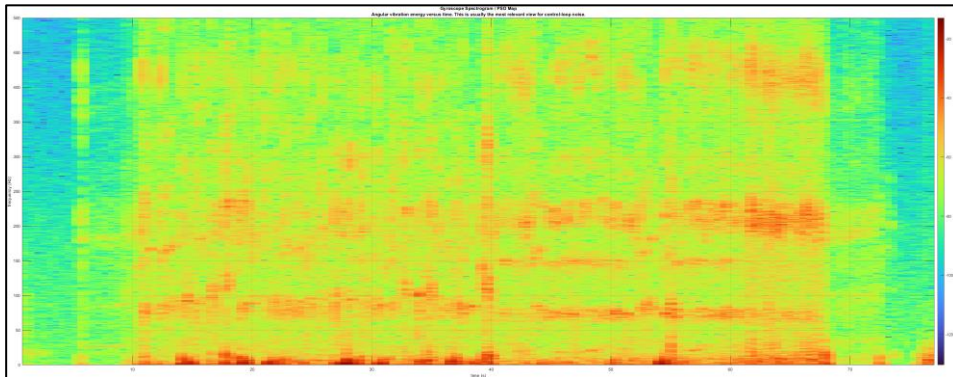
Key observation

Spectrogram hot pixels track the overlay lines, so the dominant gyro vibration is motor-driven — exactly what a dynamic notch can attenuate.

Top: Gyro spectrogram

Bottom: spectrogram + motor RPM overlay (f_0 solid, $2f_0$ / $3f_0$ dashed)

Accelerometer Spectrogram with Live RPM Overlay



Why the accelerometer?

Reflects linear airframe motion — what mechanically excites the structure, not what the PID sees.

What it diagnoses

Frame resonance — modes that amplify motor energy.

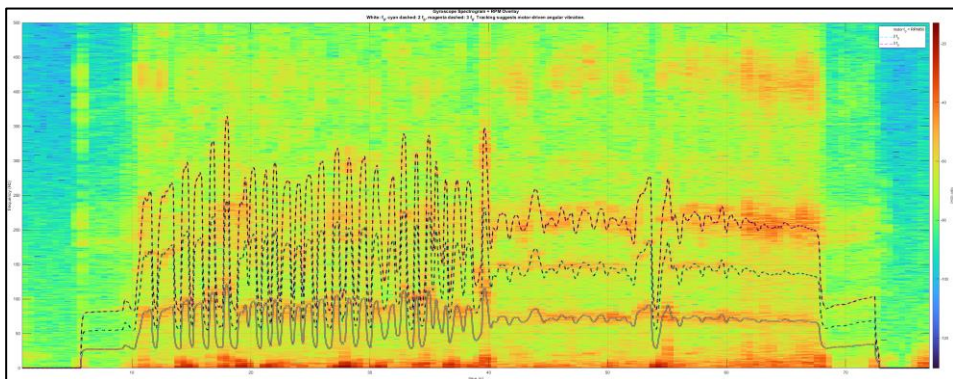
Propeller imbalance — strong f_0 relative to $2f_0 / 3f_0$.

Motor mount stiffness — broadband vs tonal.

Why both

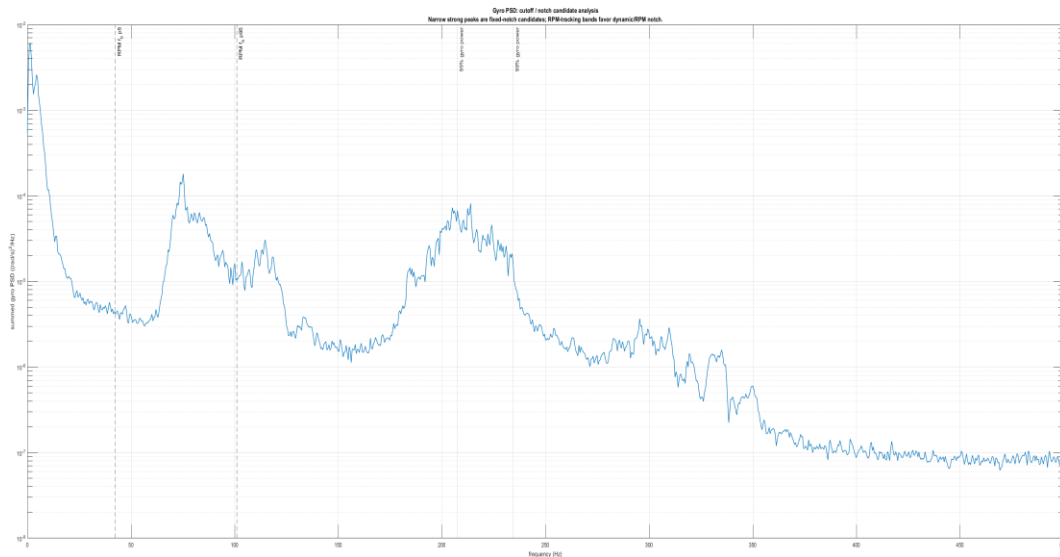
Gyro guides filter design; accel guides mechanical changes (soft mount, prop balance, frame stiffening).

Top: Accelerometer spectrogram



Bottom: spectrogram + motor RPM overlay (f_0 solid, $2f_0 / 3f_0$ dashed)

Gyro PSD — Where to Place the Filters



How to read this plot

- Frequency up to Nyquist (500 Hz at 1 kHz sampling).
- Log-scale gyro power on Y axis.
- Dashed lines: motor f_0 band (RPM p5–p95).
- Dotted lines: 95% / 99% cumulative gyro power.

What the spectrum shows

- Broad peak at 70–90 Hz — inside the RPM band, moves with throttle.
- Cluster at 180–230 Hz — blade-pass and 2nd harmonics.
- Above ~250 Hz: noise floor.

Filter design implication

- Inside the RPM band → dynamic (RPM-tracking) notch.
- Narrow stationary peaks outside it → fixed notch.

A static notch tuned to mean RPM would only catch part of this band — an RPM-tracking notch is required.